**Ramakrishna Mission Residential College (Autonomous), Narendrapur**

**Name: Soham Sarkar**

**Registration Number: A03-1112-0207-21**

**Roll Number: 6R24STSA2009**

**Semester: SIX**

**PAPER: DS41**

**Project Topic: A Study under Cauchy-Gaussian Mixture Model**

# A Study Under Cauchy-Gaussian Mixture Model

**ABSTRACT**

The article delves into the comparison of traditional parametric and nonparametric approaches with newly proposed test procedures for assessing location parameters in Cauchy-Gaussian Mixture Models (CGMMs). Through extensive simulation studies, the article introduces novel test methodologies and evaluates their efficacy against established techniques. The asymptotic null distribution of the proposed statistics are obtained and asymptotically distribution free tests are provided. We evaluate the empirical level for the compatibility of the test and simulated powers (asymptotic powers) for comparison. By leveraging simulation studies, the research endeavours to enhance our understanding of the comparative performance of these methodologies under diverse conditions. Such investigations are crucial for refining statistical inference methods, ensuring their robustness and applicability in real-world scenarios.

**KEYWORDS:**

Asymptotic distribution, Empirical distribution, Robust Statistic.

# 1. Introduction:

The basic purpose of statistical inference is to infer about some characteristics of the associated distribution on the basis of a dataset. Depending on the nature of the data, we assume a probability distribution of the concerned variable and develop theories on the basis of that assumption. Sometimes, we get such data where the data itself guides us to assume that the data is contaminated. In those situations, it is natural to assume that the observed data has generated from more than one distribution i.e., the distribution of the associated random variable can be represented by a mixture of two or more distributions rather a single one. This situation arises the concept of mixture modelling in statistics. Mixture modelling is a broad class of statistical models used to discern observed classes or patterns of responses from data. Mixture models typically represent non-normal data with a combination of normal distributions; however, combinations of other distributions can be used as well. Now we will discuss the problems associated with mixture modelling with the help of an illustrative example of Normal-Cauchy mixture model.

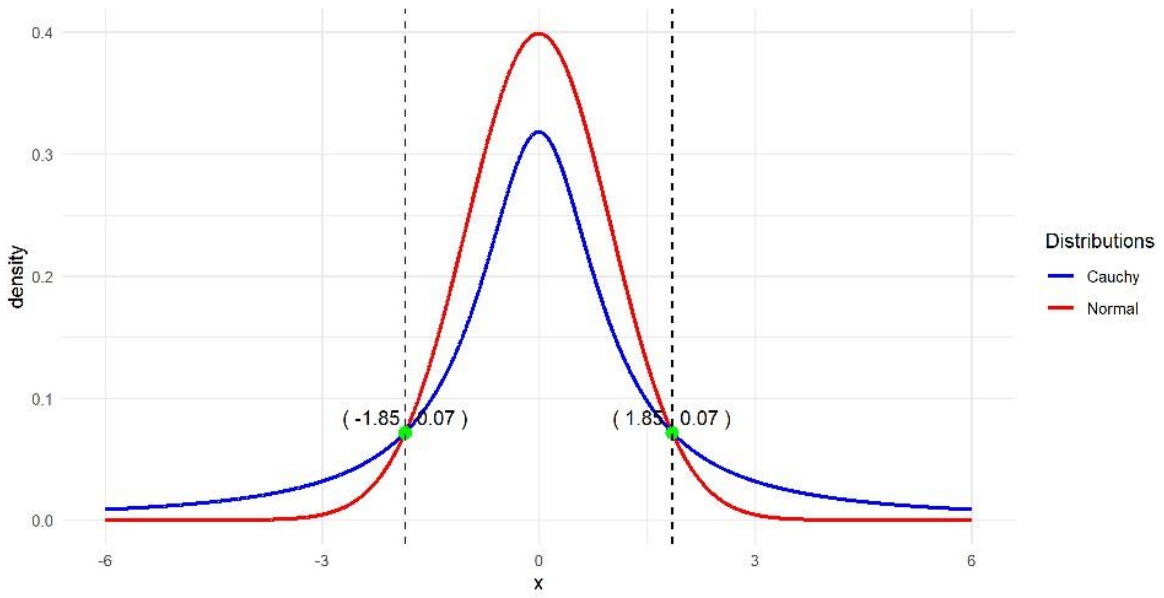



**Fig.1.1: N (0,1) & C (0,1) Mixture Distribution**

In the above figure we are considering a mixture of two symmetric distributions (Standard Normal and Standard Cauchy). We know that natural tolerance limit of Normal distribution is $(-3, 3)$ i.e., if the data has generated from Normal distribution only then it is expected that at least 99.73% of the data will generate from this interval $(-3, 3)$. Now if there is sufficient information to believe that the data contains impurities then we have some non-normal observations in our data. In this article we assume Cauchy distribution that generates impurities in our data and for natural response we use Normal distribution. But the separation of normal and non-normal observations is difficult in practice. It is clear that if an observation lies outside the interval $(-3, 3)$ it is likely to be generated from Cauchy distribution. But when an observation falls in the interval $(-1.85, 1.85)$, there is a high chance that it generates from Normal distribution (93.57%)

or Cauchy distribution (68.45%) [see Fig 1.1] and thus that observation may not be treated as an impure. However, the region $(-3,-1.85) \cup (1.85,3)$ shows challenge for discrimination between the distributions. Thus, the data has directed us to investigate different existing statistical inferences as well as it motivated us to construct new methods by which we can handle such data. We consider the following motivating example and highlight some challenges associated with the data.

## Motivating Example:

1) Noise in any communication medium plays a major role in statistical characteristics are required for data recovery; says Bayesian theory of detection. Underwater acoustic noise is dominated by impulsive sources in shallow water areas. This leads to a situation where the noise density function shows a heavier tail than that Gaussian density and the performance of traditional gaussian receiver becomes sub-optimal or even worse. Several Literatures have shown that heavy-tailed impulsive noise can be very well modelled by the bi-parameter Cauchy-Gaussian mixture distribution which is an approximation to the symmetric α-stable classes.
2) In financial market when estimating Value at Risk; the asset returns are assumed to have Gaussian Distribution. Since Gaussian distribution has a relatively thin tail, this method can substantially underestimate the market risk, especially when a crisis occurs. To overcome this issue, an alternative distribution for the asset returns should be considered. Distributions with heavy-tailed are one of those options, such as student's t, Gumbel, Levy and Cauchy distribution.

In this article, we will explore testing the average signal (stated in example 1) rather than estimating them, with a fixed scale parameter, using both parametric and nonparametric setups. Additionally, we aim to evaluate the impurities present in the signal. Parametric methods involve assuming specific distributions for the data, such as the Cauchy-Gaussian Mixture Distribution, and testing hypotheses about the means. Conversely, nonparametric methods refrain from making strong assumptions about the underlying distribution and instead rely on the data itself to conduct tests. Assessing impurities in the signal involves various approaches depending on the nature of the impurities. For instance, if outliers are suspected, robust statistical methods or outlier detection algorithms could be used.

## 2. Model Set-Up and Hypothesis:

Let the response corresponding to the $j$-th unit be denoted by $Y_j$ which is assumed to follow a continuous distribution. That is, we assume,

$$Y_j \sim iid\ F, \qquad j = 1, 2, \dots, n,$$

where $F$ is a distribution function of a continuous random variable. In this article we assume that $F$ is a mixture of $k$-continuous distribution functions i.e.

$$F = \gamma_1 F_1 + \gamma_2 F_2 + \cdots + \gamma_k F_k$$

with the restriction that $\sum_{i=1}^{k} \gamma_i = 1$ and $k < n$.

To begin with, we restrict ourselves to a mixture of two distributions only i.e., $k = 2$ and thus our model becomes,

$$F = \gamma F_1 + (1-\gamma)F_2\,, \qquad 0 < \gamma < 1$$

From the motivating examples, we have mentioned in Subsection 1.1, it is readily understood that the amount of contamination in the data is not very large. So, in our study we allow a maximum of 20% contaminated response in the sample. We further assume that $F_1 \equiv N(\mu, \sigma^2)$ and $F_2 \equiv C(\mu, \sigma)$ with the unknown location parameter $\mu \in \mathbb{R}$ and scale parameter $\sigma > 0$.

Here we focus on the general testing problem as,

$$H_0: \mu = 0 \;\; vs. \;\; H_1: \mu \neq 0.$$

Here we restrict ourselves to Cauchy-Gaussian Mixture models. In this article our objective would be

a) For a given level of significance we will study the robustness of the Student's t-statistics for varying mixing parameter and varying location and scale parameters.
b) For a given level of significance and for a fixed scale parameter (in this article we consider only $\sigma = 1$) by varying mixing parameter and location parameter we will analysis the powers using Wilcoxon Signed Rank Statistic and compare these powers with the powers obtained in part (a).
c) We can find the asymptotic cut-off of the t-statistic empirically and analyze its performance.
d) We can find a robust statistic and provide an alternative test using this statistic, i.e. for this statistic also we will find large sample and small sample cutoff and analyze its performance and compare this with (c).

There may be so many other problems regarding this mixture model but we restrict ourselves to study those above problems.

In the consequent sections we discuss the tests we will use for testing $(H_0, H_1)$. In Section 4, we provide Simulation Study. Specifically, we scrutinize the efficacy of Student's t-statistic and the Wilcoxon Signed Rank Statistic in Subsection 4.1. In Subsection 4.2, we explore the performance of both the traditional t-test statistic and a newly proposed robust statistic through empirical analysis. In Section 5, we conclude about what we get throughout our study. In Section 6, We provide potential research directions not explored in this article. The article ends with an Acknowledgement, a list of References and some R codes in Appendix.

## 3. Tests:

### 3.1 Existing Tests:

#### *(a) Parametric Test:*

For the natural response i.e., response from $N(\mu, \sigma^2)$ distribution we have the UMPU test of size $\alpha$ for testing $(H_0, H_1)$, which is defined by the critical region,

$$\omega_1 = \{\boldsymbol{x}: |T_n(\boldsymbol{x})| > t_{\alpha/2;n-1}\},$$

where $T_n = \sqrt{n}\,\bar{X}/s$ with $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$ *and* $s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2$; $t_{\alpha;n}$ is the upper $\alpha$ point of $t$-distribution with $n$ degrees of freedom. The statistic $T_n$ is known as Student's $t$-statistic, we shall check robustness of this test when impurities increase.

***(b) Nonparametric Test:***

Alternatively, by setting $\sigma = 1$ i.e., by considering location family of distributions, we can provide Wilcoxon Signed-Rank Test as the underlying distribution is symmetric. The test statistic is given by

$$W_n^+ = \sum_{i=1}^{n} R_i^+ \psi(X_i),$$

where $R_i^+$ is the rank of $X_i^+$ within $\{X_1^+, X_2^+, \ldots . X_n^+\}$ with $X_i^+ \coloneqq |X_i|$ and $\psi(x) = I_x(0, \infty)$

Here we assume that $n$ is sufficiently large so that the asymptotic normality of $W_n^+$ holds. That is, using the large sample distribution of $W_n^+$ under $H_0$, we define

$$U_n^+ = \frac{W_n^+ - n(n+1)/4}{\sqrt{n(n+1)(2n+1)/24}} \xrightarrow{D} N(0,1) \;\; as \;\; n \to \infty$$

Thus, the ADF size $\alpha$ critical region based on $U_n^+$ for testing $(H_0, H_1)$ is given by

$$\omega_2 = \{\boldsymbol{x}: |U_n^+| > \tau_{\alpha/2}\}$$

## 3.2 Proposed Tests:

***(a) Based on Empirical Distribution of $T_n$:***

It is important to observed that the exact sampling distribution of the Student's $t$-statistic is difficult to obtain under $H_0$ as the underlaying distribution is not in explicit form. Furthermore, the null distribution of $T_n$ under contaminated setup will tend to deviate from $t_{n-1}$ as the impurities increase. A simple way to handle such problem is to use the simulated null distribution of $T_n$ and to provide a test rule based on the empirical cutoff point of that distribution. Thus, an alternative to the existing parametric procedure, discussed in Subsection 3.1(a), will be provided by the critical region

$$\omega_3 = \{\boldsymbol{x}: |T_n(\boldsymbol{x})| > e_{\alpha/2}^T\},$$

where $e_\alpha^T$ is the upper $\alpha$ point of the empirical distribution of $T_n$. The choice of $e_\alpha^T$ will be discussed in the following Section Simulation Study.

***(b) Based on Robust Measures**:*

Another alternative procedure can be obtained by replacing the measures associated with the statistic $T_n$ by the corresponding robust measures. Thus, we define a new statistic

$$T_n^* = \frac{\sqrt{n}\,\tilde{X}}{QD(\boldsymbol{X})},$$

where $\tilde{X}$ is the sample median and $QD(\boldsymbol{X})$ is the sample quartile deviation.

The pdf of the response variable is

$$f(x) = \gamma f_1(x) + (1-\gamma) f_2(x), \qquad x \in \mathbb{R} \ \ and \ \ \gamma \in (0,1)$$

where $f_1(x) = \frac{1}{\sigma}\phi\left(\frac{x-\mu}{\sigma}\right) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}$ is the pdf of $N(\mu, \sigma^2)$, $f_1(x) = \frac{1}{\pi}\frac{\sigma}{\{\sigma^2+(x-\mu)^2\}}$ is the pdf of $C(\mu, \sigma)$ and $\gamma$ is the mixing parameter.

Here our objective is to study the null distribution of this new statistic $T_n^*$.

We know that

$$\sqrt{n}\left(\hat{\xi}_q - \xi_q\right) \xrightarrow{D} N\left(0, \frac{pq}{f^2(\xi_q)}\right) \ \ as \ \ n \to \infty, \qquad \cdots\cdots (1)$$

where $\xi_q$ is the $q^{th}$ order population quantile of a distribution having pdf $f(.)$, $\hat{\xi}_q$ is the sample quantile of order $q$ on the basis of a sample of size $n$, $p = 1-q, 0 < p < 1$.

For notational simplification we define,

$$Q_1 = \hat{\xi}_{1/4}\,, Q_2 = \hat{\xi}_{1/2} \ \ and \ \ Q_3 = \hat{\xi}_{3/4}.$$

Thus, we have

$$\tilde{X} = Q_2 \ \ and \ \ QD(\boldsymbol{X}) = \frac{Q_3 - Q_1}{2}.$$

Now we consider the following theorem associated with the asymptotic distribution of $T_n^*$.

***Theorem:*** *For sufficiently large n, the null distribution of $T_n^*$ can be approximated by $N(0, \eta^2(\gamma))$ i.e.,*

$$T_n^* = \frac{\sqrt{n}\,\tilde{X}}{QD(\boldsymbol{X})} \ \xrightarrow{D} \ N(0, \eta^2(\gamma)) \ \ as\ n \to \infty,$$

*where* $\eta(\gamma) = \frac{1}{2\,QD\,f(0)}$ *with $QD$ is the population quartile deviation under $H_0$.*

***Proof****:* By using asymptotic distribution of sample quartile as defined in (1) we have,

$$\sqrt{n}\left(\tilde{X} - \xi_{1/2}\right) \xrightarrow{D} N\left(0, \frac{1}{4f^2(\xi_{1/2})}\right) \ \ as \ \ n \to \infty$$

$$\Rightarrow \sqrt{n}\,\tilde{X} \xrightarrow{D} N\left(0, \frac{1}{4f^2(0)}\right) \ \ as\ n \to \infty, under\ H_0. \ \cdots\cdots (2)$$

Notice that, under $H_0, f(0) = \frac{\gamma}{\sqrt{2\pi}} + \frac{1-\gamma}{\pi} = \frac{\gamma(\sqrt{\pi}-\sqrt{2})+\sqrt{2}}{\sqrt{2\pi}}$, which is a function of $\gamma$. Further, using (1) and applying the property of CAN estimator, we have

$$\sqrt{n}(\hat{\xi}_q - \xi_q) \xrightarrow{D} N\left(0, \frac{pq}{f^2(\xi_q)}\right) \quad as \;\; n \to \infty \;\; \Rightarrow \;\; \hat{\xi}_q \xrightarrow{P} \xi_q \;\; as \; n \to \infty.$$

Thus, $Q_1 \xrightarrow{P} \xi_{1/4}$ , $Q_3 \xrightarrow{P} \xi_{3/4} \;\; as \; n \to \infty \;\Rightarrow QD(\boldsymbol{X}) \xrightarrow{P} \mathcal{QD} \;\; as \; n \to \infty \;\; \cdots\cdots (3)$.

Finally, by using Slutsky's Theorem and using (2) & (3), we have

$$T_n^* = \frac{\sqrt{n}\,\tilde{X}}{QD(\boldsymbol{X})} = \frac{\sqrt{n}\,\tilde{X}}{\mathcal{QD}} \frac{\mathcal{QD}}{QD(\boldsymbol{X})} \xrightarrow{D} \;\; N(0, \eta^2(\gamma)) \;\; as \; n \to \infty.$$

Moreover, the asymptotic variance of $T_n^*$ is a function of $\gamma$ since both $f(0)$ and $\mathcal{QD}$ are functions of $\gamma$. Hence the proof.

## 4. Simulation Study:

**4.1** In this section, we will study the performances of the tests stated in the former sections using Simulation. For Simulation we will use R programming language and its packages.

For our study, we draw random samples of varying sizes and repeat this process 100,000 times to calculate the observed values of the test statistics. To initiate our comparative analysis between the t-test statistic under the mixture set-up and the non-parametric test, we establish the nominal level (α) at 0.05. To ensure robustness and rigor in our analysis, we intentionally restrict our sample sizes to 30, 50, and 70. Given that our study allows for a maximum of 20% contaminated responses in the sample, we set the range for γ to vary up to 0.80. Throughout this article, we maintain a fixed scale parameter at σ=1 to facilitate simulation and consistency in our analyses.

***Table Number: 4.1; Title: Empirical level of the t- test and Wilcoxon Signed Rank Test***

| $\gamma \downarrow$ | ***t-test*** | | | ***Wilcoxon Signed-Rank test*** | | |
|---|---|---|---|---|---|---|
| | $n = 30$ | $n = 50$ | $n = 70$ | $n = 30$ | $n = 50$ | $n = 70$ |
| 0.99 | 0.04938 | 0.04927 | 0.04823 | 0.05081 | 0.04973 | 0.04937 |
| 0.98 | 0.04735 | 0.04494 | 0.04764 | 0.05102 | 0.04938 | 0.0504 |
| 0.97 | 0.04374 | 0.04295 | 0.04196 | 0.0486 | 0.0499 | 0.04884 |
| 0.96 | 0.04804 | 0.0441 | 0.04114 | 0.04995 | 0.04901 | 0.04842 |
| 0.95 | 0.04243 | 0.04309 | 0.03968 | 0.04935 | 0.04855 | 0.04981 |
| 0.94 | 0.04267 | 0.0384 | 0.04195 | 0.04968 | 0.04896 | 0.04983 |
| 0.92 | 0.03968 | 0.03723 | 0.03515 | 0.04953 | 0.04973 | 0.04968 |
| 0.90 | 0.03944 | 0.03969 | 0.03327 | 0.05021 | 0.04911 | 0.04896 |
| 0.88 | 0.03651 | 0.03571 | 0.03433 | 0.04905 | 0.04953 | 0.04994 |
| 0.85 | 0.0357 | 0.03256 | 0.03114 | 0.05105 | 0.04867 | 0.04911 |
| 0.82 | 0.03247 | 0.03099 | 0.02721 | 0.04995 | 0.04914 | 0.04935 |
| 0.80 | 0.03216 | 0.02754 | 0.03011 | 0.04958 | 0.04974 | 0.04969 |

In the Table 4.1, we observe a trend while using the t-test statistic: as the sample size increases along the rows, the empirical level decreases. This suggests that with larger sample sizes, there's a heightened probability of encountering contaminated data,

possibly originating from a Cauchy distribution, leading to a slight decrease in the empirical level. Similarly, along the columns, as we increase the value of the mixing parameter (γ), the empirical level decreases as expected. An increase in γ signifies a greater likelihood of encountering contaminated data, consequently lowering the empirical level. When comparing the nonparametric test to the t-test statistic based on the results in the table, it's evident that the nonparametric test yields more stable outcomes. Although the empirical level decreases with increasing sample size and mixing parameter, the rate of decrease is notably smaller compared to the t-test statistic. For instance, with γ=0.80 and n=30, the empirical level using the t-test statistic is 0.03216, whereas with the Wilcoxon Signed-Rank Statistic, it is 0.04958. This implies that the nonparametric approach results in empirical levels less influenced by the mixing parameter's value. Even with a small mixing parameter, the empirical level remains closer to the nominal level when employing the nonparametric approach. Now, let's delve into a study of the powers of the Wilcoxon Signed-Rank Test and the t-test.

***Table no: 4.2; Title: The simulated powers of the t-test and Wilcoxon Signed Rank Test***

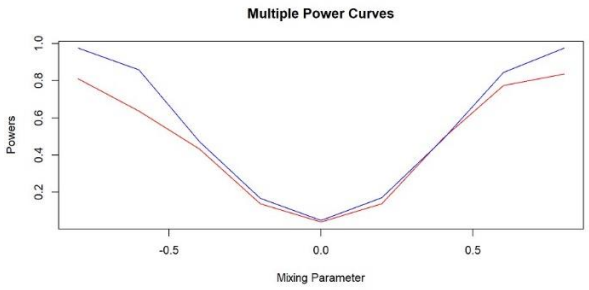
| **Sample Size (n)** | $\gamma \rightarrow$ $\mu \downarrow$ | **1.0** | **0.98** | **0.96** | **0.94** | **0.92** | **0.90** | **0.86** | **0.84** |
|---|---|---|---|---|---|---|---|---|---|
| $n = 30$ | -0.8 | 0.9884 | 0.9749 | 0.9615 | 0.8699 | 0.8108 | 0.8224 | 0.7741 | 0.6909 |
| | | (0.9848) | (0.9801) | (0.9855) | (0.9801) | (0.9754) | (0.9853) | (0.9691) | (0.9692) |
| | -0.6 | 0.8894 | 0.8228 | 0.8374 | 0.8060 | 0.6372 | 0.7450 | 0.5505 | 0.5599 |
| | | (0.8734) | (0.8744) | (0.8587) | (0.8287) | (0.8598) | (0.8140) | (0.7798) | (0.7955) |
| | -0.4 | 0.5634 | 0.5374 | 0.4865 | 0.4577 | 0.4334 | 0.3976 | 0.3662 | 0.3535 |
| | | (0.5448) | (0.5325) | (0.5133) | (0.5439) | (0.4715) | (0.4975) | (0.4985) | (0.5002) |
| | -0.2 | 0.1857 | 0.1797 | 0.1565 | 0.1563 | 0.1386 | 0.1110 | 0.1142 | 0.1148 |
| | | (0.1810) | (0.1730) | (0.1717) | (0.1716) | (0.1685) | (0.1651) | (0.1557) | (0.1590) |
| | 0.2 | 0.1840 | 0.1724 | 0.1586 | 0.1440 | 0.1380 | 0.1322 | 0.1164 | 0.1023 |
| | | (0.1796) | (0.1707) | (0.1746) | (0.1697) | (0.1706) | (0.1675) | (0.1669) | (0.1625) |
| | 0.4 | 0.5630 | 0.5076 | 0.4856 | 0.4785 | 0.4884 | 0.4993 | 0.4180 | 0.3454 |
| | | (0.5481) | (0.5302) | (0.5253) | (0.5293) | (0.4822) | (0.5116) | (0.4690) | (0.4983) |
| | 0.6 | 0.8891 | 0.8534 | 0.7786 | 0.7617 | 0.7754 | 0.6702 | 0.6166 | 0.4991 |
| | | (0.8721) | (0.8602) | (0.8440) | (0.8744) | (0.8448) | (0.8460) | (0.7955) | (0.7949) |
| | 0.8 | 0.9892 | 0.9617 | 0.9479 | 0.8262 | 0.8371 | 0.8232 | 0.7292 | 0.7382 |
| | | (0.9856) | (0.9811) | (0.9853) | (0.9697) | (0.9752) | (0.9629) | (0.9685) | (0.9439) |
| $n = 50$ | -0.8 | 0.9998 | 0.9642 | 0.9346 | 0.8726 | 0.8570 | 0.8773 | 0.7611 | 0.7131 |
| | | (0.9997) | (0.9993) | (0.9995) | (0.9993) | (0.9992) | (0.9992) | (0.9973) | (0.9985) |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | -0.6 | 0.9861 | 0.9213 | 0.8612 | 0.8500 | 0.7297 | 0.7223 | 0.7038 | 0.6278 |
| | | (0.9825) | (0.9804) | (0.9820) | (0.9796) | (0.9667) | (0.9777) | (0.9637) | (0.9556) |
| | -0.4 | 0.7897 | 0.7781 | 0.6565 | 0.6042 | 0.5667 | 0.5154 | 0.4392 | 0.4192 |
| | | (0.7752) | (0.7623) | (0.7526) | (0.7625) | (0.7125) | (0.7215) | (0.7074) | (0.6327) |
| | -0.2 | 0.2854 | 0.2622 | 0.2452 | 0.1959 | 0.2061 | 0.1711 | 0.1713 | 0.1266 |
| | | (0.2716) | (0.2722) | (0.2623) | (0.2640) | (0.2686) | (0.2491) | (0.2279) | (0.2325) |
| | 0.2 | 0.2849 | 0.2330 | 0.2563 | 0.1908 | 0.1946 | 0.1816 | 0.1480 | 0.1471 |
| | | (0.2721) | (0.2713) | (0.2646) | (0.2646) | (0.2380) | (0.2554) | (0.2161) | (0.2455) |
| | 0.4 | 0.7928 | 0.6890 | 0.6373 | 0.6387 | 0.5830 | 0.4838 | 0.4803 | 0.3988 |
| | | (0.7722) | (0.7642) | (0.7741) | (0.7328) | (0.7122) | (0.7629) | (0.6516) | (0.7159) |
| | 0.6 | 0.9861 | 0.9214 | 0.9219 | 0.7951 | 0.8568 | 0.7338 | 0.6620 | 0.5914 |
| | | (0.9821) | (0.9801) | (0.9746) | (0.9682) | (0.9796) | (0.9635) | (0.9467) | (0.9712) |
| | 0.8 | 0.9998 | 0.9787 | 0.9411 | 0.9072 | 0.8694 | 0.8017 | 0.7841 | 0.7112 |
| | | (0.9996) | (0.9993) | (0.9996) | (0.9996) | (0.9994) | (0.9990) | (0.9977) | (0.9938) |
| $\boldsymbol{n = 70}$ | -0.8 | 1.0000 | 0.9654 | 0.9323 | 0.9061 | 0.8490 | 0.7972 | 0.7805 | 0.7641 |
| | | (1.0000) | (1.0000) | (1.0000) | (1.0000) | (1.0000) | (1.0000) | (1.0000) | (1.0000) |
| | -0.6 | 0.9988 | 0.9548 | 0.8943 | 0.8132 | 0.7374 | 0.7506 | 0.7238 | 0.6966 |
| | | (0.9980) | (0.9981) | (0.9974) | (0.9965) | (0.9971) | (0.9960) | (0.9900) | (0.9952) |
| | -0.4 | 0.9104 | 0.8630 | 0.7090 | 0.7630 | 0.6655 | 0.6091 | 0.4927 | 0.4375 |
| | | (0.8969) | (0.8919) | (0.8804) | (0.8537) | (0.8646) | (0.8706) | (0.8290) | (0.8376) |
| | -0.2 | 0.3783 | 0.3408 | 0.3165 | 0.2919 | 0.2400 | 0.2071 | 0.2235 | 0.1714 |
| | | (0.3652) | (0.3534) | (0.3504) | (0.3534) | (0.3275) | (0.3313) | (0.3303) | (0.2912) |
| | 0.2 | 0.3782 | 0.3426 | 0.2724 | 0.2883 | 0.2364 | 0.2093 | 0.1665 | 0.1629 |
| | | (0.3617) | (0.3616) | (0.3535) | (0.3561) | (0.3238) | (0.3258) | (0.3241) | (0.3317) |
| | 0.4 | 0.9111 | 0.8178 | 0.7103 | 0.6944 | 0.7167 | 0.5691 | 0.4860 | 0.4677 |
| | | (0.8975) | (0.8742) | (0.8779) | (0.8863) | (0.8650) | (0.8655) | (8328) | (0.8314) |
| | 0.6 | 0.9988 | 0.9634 | 0.9188 | 0.8657 | 0.7822 | 0.7871 | 0.6333 | 0.5951 |
| | | (0.9980) | (0.9981) | (0.9972) | (0.9960) | (0.9970) | (0.9961) | (0.9948) | (0.9936) |
| | 0.8 | 1.0000 | 0.9657 | 0.9316 | 0.8932 | 0.8601 | 0.8520 | 0.7425 | 0.7566 |
| | | (1.0000) | (1.0000) | (1.0000) | (1.0000) | (1.0000) | (1.0000) | (1.0000) | (1.0000) |

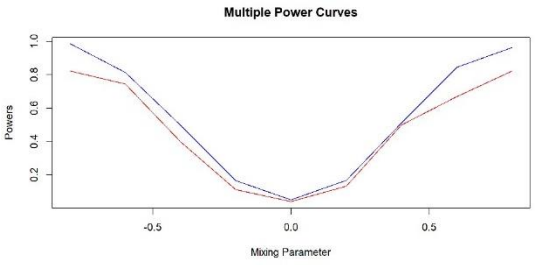
In the Table 4.2, the values without braces represent the powers of the t-test, while those within braces denote the powers of the nonparametric Wilcoxon Signed Rank Test. It's evident that as the parameter γ decreases from 1 to 0.84 while keeping μ fixed, the

powers of both tests decline. This phenomenon can be attributed to the decreasing value of γ, indicating a higher likelihood of data originating from a Cauchy Population and thus an increased chance of encountering contaminated data. Consequently, the powers of both tests diminish. Of noteworthy significance is the observation that, across all sample sizes, the powers of the parametric test exhibit some instability, whereas the nonparametric test yields more consistent results. Moreover, the nonparametric test demonstrates slightly superior power compared to its parametric counterpart. When γ remains fixed and μ varies from -0.8 to 0.8, the nonparametric test consistently outperforms the parametric test in terms of both stability and power. This suggests that in scenarios where data contamination is a concern or when the assumptions of parametric tests are questionable, opting for nonparametric methods such as the Wilcoxon Signed-Rank Test can offer more reliable and robust results.

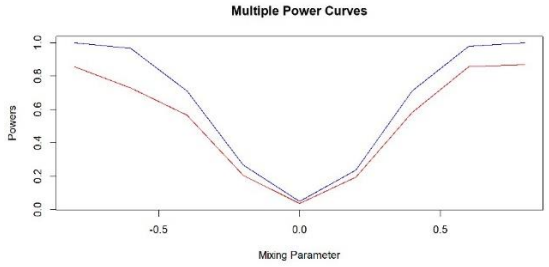
We'll now delve into the power curves of these two tests to conduct a graphical comparison of their performance. By plotting these curves, we can visually assess how the parametric (t-test) and nonparametric (Wilcoxon Signed-Rank Test) methods fare across different scenarios. Through these graphical representations, we aim to gain a deeper understanding of how varying sample sizes, mixing parameters (γ), and mean (μ) values influence the performance of each test. This visual analysis will offer valuable insights into the comparative effectiveness of these statistical approaches.



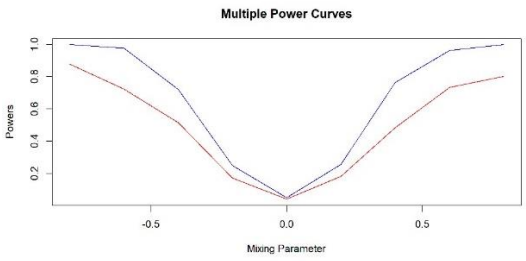

**n=30, $\gamma = 0.92$**



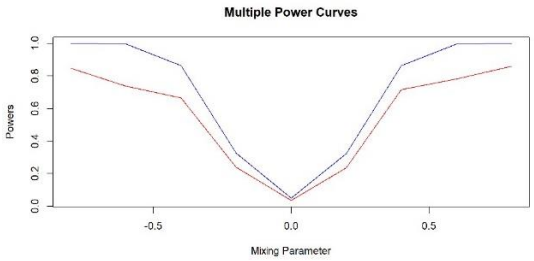

**n=30, $\gamma = 0.90$**

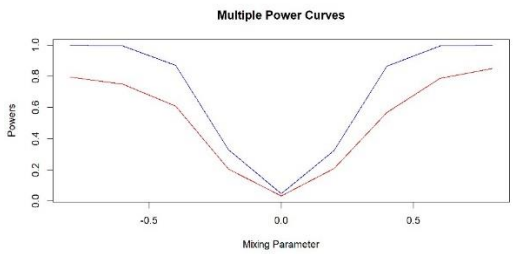



**n=50, $\gamma = 0.92$**




**n=50, $\gamma = 0.90$**




**n=70, $\gamma = 0.92$** **n=70, $\gamma = 0.90$**

**Figure Description:** The Blue Line Diagram Shows the Powers of the Wilcoxon Signed Rank Test and The Red Line Diagram Shows the Powers of the t-test.

***NOTE:*** As we increase the chance of getting contaminated data either by increase in the sample size or by increase the mixing-parameter we start to get unstable data which is quite obvious. So, in the article to enhance the stability and reliability of our data, we've undertaken 20,000 simulations, each iterated five times and taken the average value thereafter for each fixed sample size and fixed mixing-parameter. This meticulous method allows for a comprehensive exploration of the dataset's variability and consistency. By conducting multiple simulations, we aim to encompass a wide range of potential outcomes while minimizing the influence of random fluctuations. Such rigorous methodology not only strengthens the robustness of our analysis but also fortifies the credibility of our findings.

**4.2** In this section, as previously mentioned **Proposed Tests Section**, our focus shifts to examining the empirical behavior of the t-test statistic under the mixture set-up particularly concerning our hypothesis of interest $H_0: \mu = 0\ vs. H_1: \mu \neq 0$ ,given the potential deviation of the statistic T from the t-distribution under these conditions, we resort to simulation to ascertain empirical cut-offs at the 5% level for different sample sizes. The findings of the analysis are detailed in the table below:

***Table no: 4.3; Title: The Empirical cut-off of the test using t-test statistic***

| $\gamma \downarrow$ | n=30 | n=50 | n=70 | n=100 | n=150 | n=200 | n=500 |
|---|---|---|---|---|---|---|---|
| 0.99 | 1.696914 | 1.636791 | 1.631634 | 1.637855 | 1.613636 | 1.633072 | 1.628357 |
| 0.98 | 1.690613 | 1.655885 | 1.654589 | 1.598095 | 1.640815 | 1.603434 | 1.560023 |
| 0.97 | 1.662383 | 1.601731 | 1.614087 | 1.629031 | 1.584908 | 1.585155 | 1.583198 |
| 0.96 | 1.693298 | 1.648948 | 1.639064 | 1.609699 | 1.572683 | 1.555074 | 1.540765 |
| 0.95 | 1.637302 | 1.603926 | 1.651756 | 1.563839 | 1.615939 | 1.56419 | 1.534107 |
| 0.94 | 1.681724 | 1.590925 | 1.589853 | 1.584451 | 1.559055 | 1.550005 | 1.550019 |
| 0.93 | 1.656157 | 1.599914 | 1.588516 | 1.563348 | 1.535753 | 1.567957 | 1.535453 |
| 0.92 | 1.608472 | 1.604026 | 1.581088 | 1.530927 | 1.556084 | 1.544541 | 1.516376 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 0.90 | 1.588209 | 1.554098 | 1.59316 | 1.565276 | 1.572929 | 1.525391 | 1.510911 |
| 0.88 | 1.620512 | 1.569247 | 1.568168 | 1.564457 | 1.527008 | 1.526995 | 1.51274 |
| 0.85 | 1.631004 | 1.563502 | 1.55883 | 1.538967 | 1.519437 | 1.505162 | 1.508831 |
| 0.82 | 1.587799 | 1.541449 | 1.528166 | 1.521185 | 1.517383 | 1.51977 | 1.50914 |
| 0.80 | 1.604074 | 1.561712 | 1.522195 | 1.514904 | 1.523803 | 1.50833 | 1.504497 |

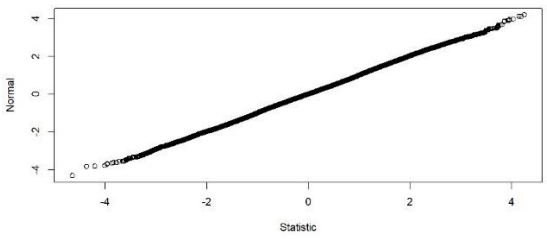
The Table 4.3 illustrates that for a fixed γ, the values tend to stabilize as the sample size (n) increases. This behaviour is expected, as larger sample sizes lead to the convergence of the distribution of any statistic, suitably transformed, towards N(0,1). To ensure stability in determining the empirical level, we select a moderate sample size, such as n=100, to establish cutoff values. Averaging these cutoff values enhances stability. Subsequently, using this averaged cutoff point, we calculate the empirical level, as presented in Table 4.4.

***Table no: 4.4; Title: The Empirical level of the test using t-test statistic***

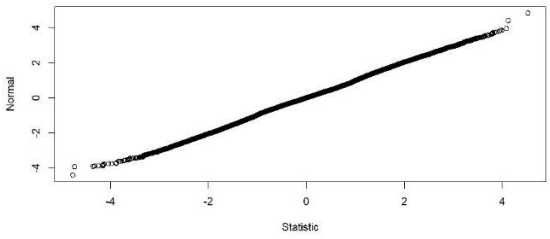
| $\gamma \downarrow$ | n=30 | n=50 | n=70 |
|---|---|---|---|
| 0.99 | 0.06087 | 0.06054 | 0.05932 |
| 0.98 | 0.06233 | 0.05984 | 0.05718 |
| 0.97 | 0.06109 | 0.05765 | 0.05516 |
| 0.96 | 0.05721 | 0.05726 | 0.05373 |
| 0.95 | 0.05861 | 0.05776 | 0.05599 |
| 0.94 | 0.06015 | 0.05459 | 0.05132 |
| 0.93 | 0.05976 | 0.053 | 0.05122 |
| 0.92 | 0.05532 | 0.05164 | 0.05263 |
| 0.90 | 0.05517 | 0.05034 | 0.04924 |
| 0.88 | 0.05286 | 0.05216 | 0.04997 |
| 0.85 | 0.05072 | 0.05014 | 0.04708 |
| 0.82 | 0.04935 | 0.0463 | 0.04677 |
| 0.80 | 0.05157 | 0.04912 | 0.04545 |

It's clearly discernible from the Table 4.4 that our empirically derived levels closely mirror our predetermined nominal levels. This alignment between empirical and nominal levels underscores the robustness of our methodology. A notable trend emerges as we delve deeper into the data: as the sample size increases, holding γ constant, there's a discernible increase in the likelihood of data originating from a Cauchy Distribution. This phenomenon inherently diminishes the empirical level, indicating a higher probability of encountering outliers or extreme values in our dataset. Similarly, when we maintain a fixed sample size and increase the mixing parameter γ, we witness a parallel rise in the likelihood of data exhibiting characteristics of a Cauchy Distribution. This progressive shift towards data with heavy tails contributes to the decline in the empirical level. These findings underscore the intricate interplay between sample size and mixing parameters in shaping the distributional properties of our dataset. Understanding these dynamics is crucial for robust statistical inference, especially in scenarios where non-normality or outliers may significantly impact the validity of our conclusions. Moving

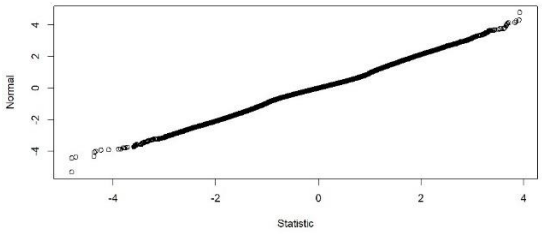
forward, our analysis will delve into calculating the simulated empirical power of our tests. However, before delving into these calculations, it's imperative to gain deeper insights into the behaviour of the t-test statistic as the sample size varies. To achieve this, we will examine q-q plots of the distribution of the t-test statistic. These plots will offer valuable visual cues to discern any discernible patterns or deviations from normality as sample size increases. Such insights will enrich our understanding of the statistical properties underlying our dataset and enhance the robustness of our subsequent analyses.



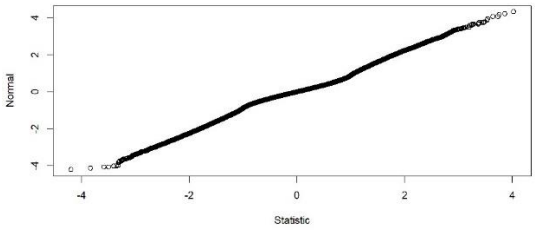

**n=50, $\gamma = 0.94$**




**n=70, $\gamma = 0.94$**




**n=100, $\gamma = 0.94$**




**n=200, $\gamma = 0.94$**

From these q-q plots we can clearly see that as sample size increases the distribution of t-statistics get stabilized.

Now, we turn our attention to comparing the performance of the aforementioned statistic with another proposed statistic, denoted as $T_n^*$, as outlined in the preceding section on Proposed Tests. Employing simulation once again, we endeavour to determine empirical cut-offs at the 5% significance level across varying sample sizes. The results of this investigation are presented in the table below:

***Table no: 4.5; Title: The Empirical cut-off of the $T_n^*$ test(small sample size)***

| $\gamma \downarrow$ | n=30 | n=50 | n=70 |
|---|---|---|---|
| 0.99 | 3.280832 | 3.183349 | 3.156992 |
| 0.98 | 3.294918 | 3.184249 | 3.139578 |
| 0.97 | 3.256816 | 3.171586 | 3.154825 |
| 0.96 | 3.265444 | 3.1445 | 3.142062 |
| 0.95 | 3.27696 | 3.152184 | 3.137909 |

| | | | |
|---|---|---|---|
| 0.94 | 3.275977 | 3.149106 | 3.140637 |
| 0.93 | 3.231656 | 3.178365 | 3.128586 |
| 0.92 | 3.251387 | 3.157755 | 3.103341 |
| 0.90 | 3.266404 | 3.188266 | 3.116312 |
| 0.88 | 3.153365 | 3.155269 | 3.091276 |
| 0.85 | 3.247419 | 3.131677 | 3.090312 |
| 0.82 | 3.240398 | 3.133983 | 3.084903 |
| 0.80 | 3.192677 | 3.135547 | 3.095403 |

***Table no: 4.6; Title: The Empirical cut-off of the $T_n^*$ test (large sample size)***

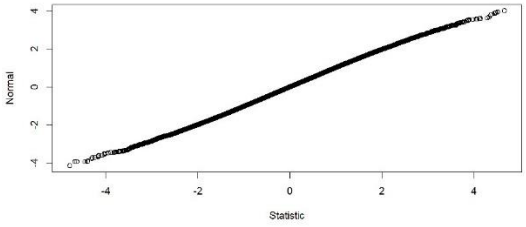
| $\gamma \downarrow$ | **n=100** | **n=200** | **n=500** |
|---|---|---|---|
| 0.99 | 3.10825 | 3.083392 | 3.061715 |
| 0.98 | 3.105972 | 3.079204 | 3.081089 |
| 0.97 | 3.063439 | 3.068974 | 3.053803 |
| 0.96 | 3.109142 | 3.042975 | 3.08313 |
| 0.95 | 3.09325 | 3.092409 | 3.033709 |
| 0.94 | 3.119668 | 3.08729 | 3.035504 |
| 0.93 | 3.096088 | 3.092835 | 3.049532 |
| 0.92 | 3.120916 | 3.057703 | 3.038883 |
| 0.90 | 3.08978 | 3.061971 | 3.036029 |
| 0.88 | 3.066178 | 3.060488 | 3.024719 |
| 0.85 | 3.102684 | 3.057123 | 3.020198 |
| 0.82 | 3.074714 | 3.035417 | 3.01047 |
| 0.80 | 3.068945 | 3.046405 | 3.019115 |

This comparative analysis aims to provide insights into the effectiveness of $T_n^*$ relative to the previously discussed statistic. By examining empirical cut-offs across different sample sizes, we can assess the robustness and reliability of $T_n^*$ in accurately capturing the underlying distributional properties of the data. This exploration serves as a crucial step in our quest to identify the most suitable statistical approach for our analysis.

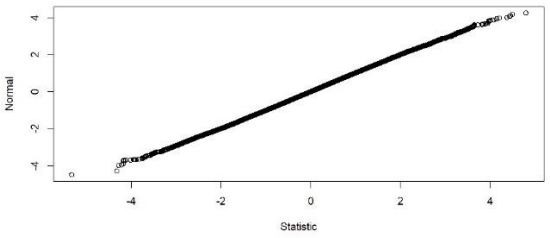
After closely examining the tables presented above, a discernible trend emerges: as the sample size (n) increases while holding γ constant, there's a notable convergence of values towards stability. This phenomenon is a direct consequence of the well-understood statistical principle that with larger sample sizes, the distribution of any statistic, given a suitable transformation, tends to converge towards the standard normal distribution N(0,1). This convergence towards normality is a hallmark of statistical inference, bolstering the reliability of our analyses. In our quest to determine the empirical level under these conditions, we opt for a sample size of n=50. This choice strikes a balance between computational efficiency and statistical robustness. By computing cutoff values at this moderate sample size and subsequently averaging them, we arrive at a more stable cutoff point. This averaged cutoff point serves as our anchor for evaluating simulated power, a critical metric in assessing the efficacy of our statistical tests. However, before delving into the intricacies of simulated power calculations, we embark on a preliminary exploration of the dataset's behaviour as sample size increases. To accomplish this, we turn our attention to q-q plots, powerful graphical tools used to assess the distributional properties of data. These plots provide a visual representation

of the quantile-quantile relationship between our dataset and a standard normal distribution. By scrutinizing these q-q plots, we aim to discern any deviations from normality and gain insights into how the statistical properties of our dataset evolve with varying sample sizes. By meticulously analysing q-q plots and understanding their implications, we lay a robust foundation for our subsequent analyses. Armed with this deeper understanding, we proceed to calculate simulated power, comparing it with the powers of previously proposed statistics. This comprehensive approach ensures that our statistical conclusions are not only reliable but also grounded in a thorough understanding of the underlying data dynamics.

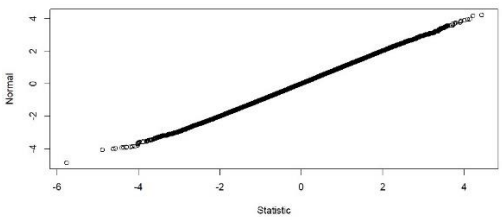
***NOTE:*** As we increase the chance of getting contaminated data either by increase in the sample size or by increase the mixing-parameter we start to get unstable data which is quite obvious. So, in the article to enhance the stability and reliability of our data, we've undertaken 20,000 simulations, each iterated five times and taken the average value thereafter for each fixed sample size and fixed mixing-parameter. This meticulous method allows for a comprehensive exploration of the dataset's variability and consistency. By conducting multiple simulations, we aim to encompass a wide range of potential outcomes while minimizing the influence of random fluctuations. Such rigorous methodology not only strengthens the robustness of our analysis but also fortifies the credibility of our findings.



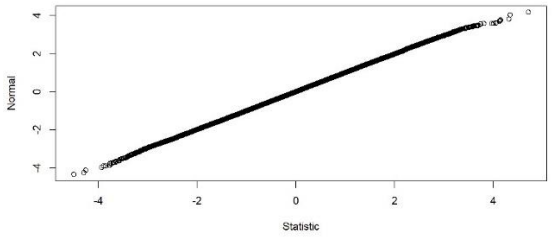

**n=50, $\gamma = 0.94$**




**n=70, $\gamma = 0.94$**




**n=100, $\gamma = 0.94$**




**n=200, $\gamma = 0.94$**

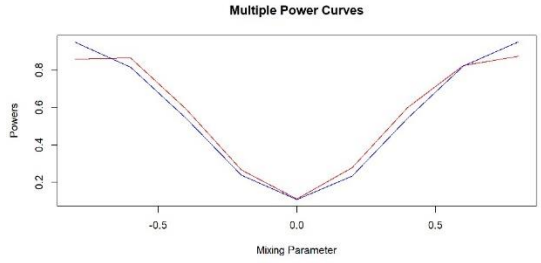
**Figure Description:** From these q-q plots we can clearly see that as sample size increases the distribution of the robust statistic $T_n^*$ gets stabilized,

***Table no: 4.7; Title: The simulated powers of $T_n$ test and $T_n^*$ test***

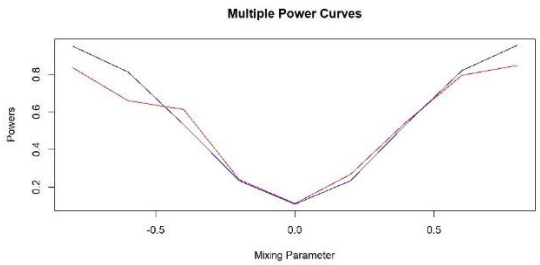
| **Sample Size (n)** | $\gamma \rightarrow$ $\mu \downarrow$ | **1.0** | **0.98** | **0.96** | **0.94** | **0.92** | **0.90** | **0.86** | **0.84** |
|---|---|---|---|---|---|---|---|---|---|
| $n = 30$ | -0.8 | 0.99707 | 0.97754 | 0.93925 | 0.86383 | 0.85863 | 0.83673 | 0.83785 | 0.77482 |
| | | (0.96216) | (0.96065) | (0.96054) | (0.95715) | (0.94939) | (0.95288) | (0.94726) | (0.94923) |
| | -0.6 | 0.95399 | 0.92679 | 0.89097 | 0.8278 | 0.8643 | 0.66152 | 0.72298 | 0.6691 |
| | | (0.83465) | (0.83298) | (0.82597) | (0.82376) | (0.81545) | (0.81408) | (0.80942) | (0.80933) |
| | -0.4 | 0.73439 | 0.69477 | 0.66906 | 0.62488 | 0.59262 | 0.61336 | 0.46915 | 0.46404 |
| | | (0.55332) | (0.54893) | (0.54722) | (0.53901) | (0.54283) | (0.53337) | (0.51363) | (0.51773) |
| | -0.2 | 0.33195 | 0.31968 | 0.28538 | 0.25935 | 0.26703 | 0.23922 | 0.26772 | 0.24072 |
| | | (0.24354) | (0.24042) | (0.23736) | (0.23835) | (0.23925) | (0.23316) | (0.23273) | (0.23569) |
| | 0.2 | 0.32961 | 0.28733 | 0.29164 | 0.29275 | 0.27978 | 0.26807 | 0.23134 | 0.23889 |
| | | (0.24569) | (0.24392) | (0.23715) | (0.23243) | (0.23344) | (0.23345) | (0.23099) | (0.22092) |
| | 0.4 | 0.73201 | 0.64176 | 0.70767 | 0.64626 | 0.60049 | 0.5463 | 0.55788 | 0.49433 |
| | | (0.55327) | (0.55256) | (0.54573) | (0.54439) | (0.54195) | (0.53572) | (0.52179) | (0.51773) |
| | 0.6 | 0.95535 | 0.95533 | 0.86299 | 0.84098 | 0.82549 | 0.79655 | 0.69721 | 0.59412 |
| | | (0.8332) | (0.82978) | (0.83207) | (0.82182) | (0.82204) | (0.82062) | (0.80674) | (0.81314) |
| | 0.8 | 0.99715 | 0.9316 | 0.9493 | 0.8818 | 0.87475 | 0.84952 | 0.79786 | 0.79856 |
| | | (0.96364) | (0.96334) | (0.95778) | (0.95783) | (0.95124) | (0.95625) | (0.95494) | (0.95037) |
| $n = 50$ | -0.8 | 0.99993 | 0.97836 | 0.91764 | 0.91393 | 0.89117 | 0.87802 | 0.79322 | 0.76876 |
| | | (0.9969) | (0.9967) | (0.99615) | (0.99575) | (0.99561) | (0.99552) | (0.99478) | (0.99457) |
| | -0.6 | 0.99626 | 0.96523 | 0.92502 | 0.88556 | 0.87207 | 0.78999 | 0.74892 | 0.78163 |
| | | (0.95036) | (0.95263) | (0.94937) | (0.95105) | (0.94237) | (0.94406) | (0.94038) | (0.93588) |
| | -0.4 | 0.89538 | 0.83373 | 0.80256 | 0.70503 | 0.67322 | 0.62796 | 0.59676 | 0.56251 |
| | | (0.72282) | (0.7211) | (0.71844) | (0.70803) | (0.70286) | (0.70146) | (0.70748) | (0.69161) |
| | -0.2 | 0.44251 | 0.41721 | 0.36641 | 0.36037 | 0.32603 | 0.30383 | 0.29258 | 0.25291 |
| | | (0.31119) | (0.30461) | (0.30556) | (0.29919) | (0.29901) | (0.30337) | (0.29066) | (0.29071) |
| | 0.2 | 0.44217 | 0.40325 | 0.40696 | 0.34646 | 0.35048 | 0.32561 | 0.26558 | 0.27926 |
| | | (0.30719) | (0.30761) | (0.30259) | (0.30134) | (0.30071) | (0.29671) | (0.29242) | (0.29331) |
| | 0.4 | 0.89376 | 0.77355 | 0.79408 | 0.77361 | 0.71037 | 0.58358 | 0.66336 | 0.55123 |
| | | (0.72292) | (0.71387) | (0.71831) | (0.70924) | (0.70935) | (0.71623) | (0.68932) | (0.69051) |
| | 0.6 | 0.99601 | 0.94746 | 0.94797 | 0.89452 | 0.85817 | 0.82475 | 0.72997 | 0.75104 |
| | | (0.9538) | (0.9508) | (0.95187) | (0.95011) | (0.94827) | (0.94602) | (0.94279) | (0.93774) |
| | 0.8 | 0.99998 | 0.97755 | 0.961 | 0.87739 | 0.90657 | 0.85241 | 0.81989 | 0.80481 |
| | | (0.99691) | (0.99683) | (0.9961) | (0.99613) | (0.99635) | (0.99539) | (0.9947) | (0.99388) |
| $n = 70$ | -0.8 | 1.0000 | 0.96145 | 0.95395 | 0.94649 | 0.8826 | 0.85253 | 0.81253 | 0.81869 |
| | | (0.99988) | (0.99975) | (0.99976) | (0.99975) | (0.99971) | (0.99969) | (0.9995) | (0.99949) |
| | -0.6 | 0.99968 | 0.967 | 0.89239 | 0.90851 | 0.85441 | 0.82741 | 0.75105 | 0.70751 |
| | | (0.98852) | (0.98777) | (0.9867) | (0.98584) | (0.98577) | (0.98446) | (0.98423) | (0.98316) |
| | -0.4 | 0.96086 | 0.88824 | 0.87979 | 0.79306 | 0.70682 | 0.72344 | 0.57592 | 0.59551 |
| | | (0.83777) | (0.8342) | (0.82858) | (0.82851) | (0.82285) | (0.82236) | (0.8079) | (0.81276) |
| | -0.2 | 0.54428 | 0.51026 | 0.45394 | 0.46211 | 0.38386 | 0.33942 | 0.32818 | 0.32376 |
| | | (0.37616) | (0.37749) | (0.36565) | (0.36858) | (0.36389) | (0.36289) | (0.35454) | (0.34938) |
| | 0.2 | 0.54173 | 0.49148 | 0.45036 | 0.46279 | 0.37779 | 0.36542 | 0.32443 | 0.25851 |
| | | (0.37439) | (0.37488) | (0.37023) | (0.36854) | (0.36312) | (0.36209) | (0.35903) | (0.34999) |
| | 0.4 | 0.96161 | 0.89783 | 0.8092 | 0.78215 | 0.76002 | 0.67534 | 0.60207 | 0.60362 |
| | | (0.83904) | (0.83364) | (0.835) | (0.82523) | (0.82281) | (0.81751) | (0.80891) | (0.81111) |
| | 0.6 | 0.99981 | 0.94124 | 0.9291 | 0.87398 | 0.86202 | 0.79141 | 0.75211 | 0.7706 |
| | | (0.9887) | (0.98761) | (0.9872) | (0.98671) | (0.98552) | (0.98475) | (0.98195) | (0.98103) |
| | 0.8 | 1.0000 | 0.97267 | 0.97069 | 0.92335 | 0.85993 | 0.85254 | 0.86776 | 0.78307 |
| | | (0.99986) | (0.99973) | (0.99977) | (0.99979) | (0.99972) | (0.99959) | (0.99961) | (0.99956) |

In the Table 4.7 the values without braces indicate the values of the simulated powers of the t-test statistic and the values with braces indicate the simulated powers of the statistic $T_n^*$. For every sample size we see that as the value of $\mu$ increases from (-0.8) to (-0.2) the powers decrease for both the statistics and as the value of $\mu$ increases from (0.2) to (0.8) the powers increase. Now we see that for almost any situations considered in our study the values of empirical powers for the $T_n^*$ are greater than the values of empirical powers for the t-test statistic. On the other hand, for a fixed $\mu$ as impurity increases the empirical powers for statistic $T_n^*$ are much more stable than the t-test statistic, this means an increase in impurity has minimal effect on this statistic $T_n^*$ compared to the t-test statistic, this is because this statistic is more robust that the t-test statistic. We know that Robust statistics are preferred when a dataset contains impurity because they are less influenced by outliers and noise. Traditional statistical methods, such as means and standard deviations, can be heavily affected by outliers or irrelevant data points present in impure datasets. This can lead to skewed results and inaccurate conclusions about the underlying data. In contrast, robust statistics are designed to be more resilient to such disturbances. They focus on measures that are less affected by extreme values, such as medians instead of means and interquartile ranges instead of standard deviations. By relying on these robust measures, analysts can obtain more accurate and reliable insights from impure datasets, mitigating the impact of outliers and noise on their analyses. Therefore, when dealing with impurity in datasets, employing robust statistics helps ensure the validity and robustness of statistical analyses.

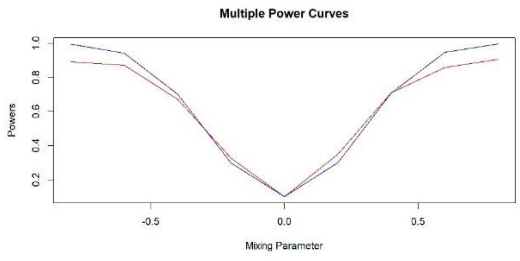
We can also see that as the impurity increases the powers decrease significantly for both the statistics which is quite obvious. Now we will study their comparison from their power curves, which are shown below.



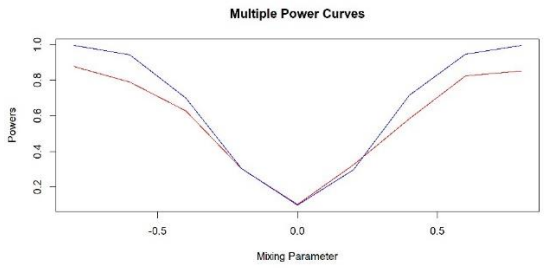

**n=30, $\gamma = 0.92$**



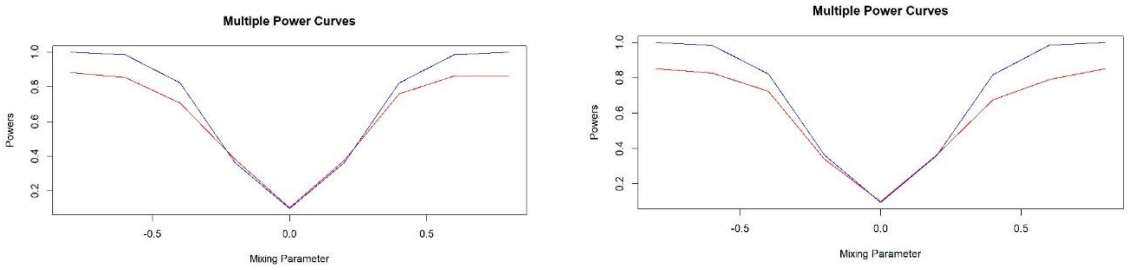

**n=30, $\gamma = 0.90$**




**n=50, $\gamma = 0.92$**




**n=50, $\gamma = 0.90$**




**n=70, $\gamma = 0.92$** **n=70, $\gamma = 0.90$**

**Figure Description:** The Blue Line Diagram Shows the Empirical Powers of the Robust statistic and The Red Line Diagram Shows the Empirical Powers of the t-test statistic.

## 5. Conclusion:

Within the scope of this article, we delve into a comprehensive simulation study to address the multifaceted objectives delineated in **Section 2**. Our meticulous exploration unfolds in **Section 4.1**, where we meticulously scrutinize the performance of two pivotal statistical tests: the nonparametric Wilcoxon Signed Rank Test and the parametric t-test. Our findings illuminate a compelling narrative: amidst escalating levels of data contamination, the nonparametric Wilcoxon Signed Rank Test emerges as a beacon of stability, steadfastly maintaining its efficacy. Notably, it exhibits a commendable increase in power compared to its parametric counterpart, the t-test. Consequently, our discerning analysis leads us to advocate for the adoption of a non-parametric setup in scenarios where a priori knowledge suggests the presence of data contamination originating from a mixture distribution. This strategic preference aligns with our overarching objective of ensuring robust and reliable statistical inference in the face of diverse data challenges. Transitioning to **Section 4.2**, we embark on an innovative exploration, introducing two novel approaches to test formulation. Our pioneering methodologies involve computing the empirical powers of both the statistic $T_n$ and the robust statistic $T_n^*$. Informed by the insights gleaned from our earlier investigation, we undertake a meticulous comparison of these two approaches. Intriguingly Our analysis indicates that the robust counterpart $T_n^*$ demonstrates notable stability in power, surpassing the empirical t-test statistic, particularly in the context of escalating levels of data contamination. This nuanced comparison underscores the nuanced interplay between statistical methodology and the inherent characteristics of the dataset. By elucidating the relative strengths and weaknesses of different testing approaches, we aim to equip researchers and practitioners with the knowledge and tools necessary to make informed decisions when conducting statistical analyses in real-world scenarios. Ultimately, our overarching objective remains steadfast: to foster a deeper understanding of statistical methodologies and their implications, thereby empowering practitioners to navigate the complexities of data analysis with confidence and precision.

## 6. Further Study:

In the article as far now we overlook two very important problems due to some unavoidable limitation which can be addressed further with more strong evidence.

(1) In Section 3, we established a crucial theorem regarding the approximate null distribution of our proposed robust statistic, $T_n^*$. A key observation from this theorem was the dependence of $T_n^*$'s approximate variance on the mixing parameter $\gamma$. However, incorporating $\gamma$ into simulation studies poses challenges due to its complexity. To address this challenge, we adopted a pragmatic approach in our simulations, using empirical cut-offs and powers to streamline our tests. While effective, this approach has its limitations as it may overlook nuances in $T_n^*$'s performance under different $\gamma$ values. A more sophisticated solution involves leveraging Bayesian Inference to incorporate prior distributions of $\gamma$ into our analyses. By doing so, we can enhance the precision and robustness of our statistical tests. Although delving into Bayesian Inference is beyond the scope of our current study, it holds promise for future research to refine and advance our methodologies. In summary, while our current approach offers practicality, the broader theoretical landscape of Bayesian methods presents opportunities for further optimization and refinement.

(2) An additional challenge arises in estimating the mixing parameter γ, which represents the proportion of impurities in the data. Commonly, the EM Algorithm is employed for this task in scenarios utilizing a mixture setup. In our study, we endeavoured to estimate γ using this technique, as detailed in Section Appendix. However, the results obtained were notably unstable and deemed unreliable for our dataset. The volatility in estimating γ presents a significant obstacle to our analysis. While the EM Algorithm is widely regarded for parameter estimation in mixture models, its efficacy can fluctuate based on data characteristics and model complexity. To overcome this challenge, we recognize the necessity for innovative approaches. In subsequent phases of our research, we intend to explore alternative methods for estimating γ that offer enhanced stability and reliability. By devising and implementing novel strategies, we aim to mitigate this limitation and bolster the robustness of our analysis. In summary, while the EM Algorithm serves as a valuable tool for parameter estimation, its limitations underscore the ongoing need for exploration and innovation in statistical methodology

## Acknowledgement:

I extend my heartfelt gratitude to all those who contributed to the successful completion of this project. I would like to express my sincere appreciation to Dr. Suryasish Chatterjee for his invaluable guidance, mentorship, and unwavering support throughout the duration of this project. I would also like to extend my sincere appreciation to Dr. Debajit Chatterjee, Assistant Professor, Department of Statistics, Asutosh College, Kolkata, for his invaluable ideas and insights contributed to this project. I am also immensely grateful to our HOD sir, Dr. Parthasarathi Chakraborty for providing me with the necessary resources and facilities to conduct my project

effectively. Finally, I extend my thanks to my family and friends for their encouragement and understanding during this journey. This project would not have been possible without the collective effort and support of all those involved, and for that, I am truly grateful.

## Reference:


(1) Underwater Acoustics Communication in the Presence of Heavy-tailed Impulsive Noise with Bi-parameter Cauchy-Gaussian Mixture Model: Sharbari Banerjee and Monika Agarwal.

(2) Gaussian-Cauchy Mixture Modelling for Robust Signal-Dependent Noise Estimation: Lucio Azzari and Alessandro Foi.

(3) Bi-parameter CGM Model for approximation of $\alpha$-stable PDF.


## Appendix

**R-Codes:**

**(1)** To draw the Mixture Distribution:

```r
library(ggplot2)
x<-seq(-6,6,length.out=10000)
df_normal<-data.frame(x=x,density=dnorm(x))
df_cauchy<-data.frame(x=x,density=dcauchy(x))
normal_pdf <- dnorm(x)
cauchy_pdf <- dcauchy(x)
intersection_indices <- which(diff(sign(normal_pdf - cauchy_pdf)) != 0)
intersection_points <- data.frame(x = x[intersection_indices],
                                  y = normal_pdf[intersection_indices])
ggplot()+
  geom_line(data=df_normal,aes(x=x,y=density,color="Normal"),size=1)+
  geom_line(data=df_cauchy,aes(x=x,y=density,color="Cauchy"),size=1)+
  geom_point(data = intersection_points, aes(x = x, y = y), color = "green", size = 3) +
  annotate("text", x = intersection_points$x, y = intersection_points$y,
           label = paste("(", round(intersection_points$x, 2), ",", round(intersection_points$y, 2), ")"),
           vjust = -0.5, color = "black") +
  geom_vline(xintercept = intersection_points$x, color = "black", linetype = "dashed") +

  scale_color_manual(values=c("blue","red"))+
  labs(color="Distributions")+
  theme_minimal()
```

**(2)** To find the Empirical Level and Power of the Student's t-test:

```r
m<-100000
n<-30
a<-0.05
g<-0.98
mu<-0
s<-1
T<-double(n)
T1<-double(n)
r<-runif(n)
sum(r<=g)
x<-c()
y<-c()
for(i in 1:m)
{
    for(j in 1:n)
    {
      if(r[j]<=g)
        x[j]<-rnorm(1,mu,s)
      else
        x[j]<-rcauchy(1,mu,s)
    }

  T[i]<-(sqrt(n)*(mean(x)))/sd(x)
}
count<-sum(abs(T)>qt(1-(a/2),df=n-1))
prob<-count/m
prob
```

**(3)** To find the Empirical Level and Power of the Wilcoxon Signed Rank Test:

```r
m<-100000
n<-30
a<-0.05
g<-0.98
r<-runif(n)
mu<-0
s<-1
x<-c()
U<-c()
W<-c()
V<-c()
M<-c()
for(i in 1:m)
{for(j in 1:n)
    {if(r[j]<=g)
      x[j]<-rnorm(1,mu,s)
    else
      x[j]<-rcauchy(1,mu,s)}
  U[i]<-sum(rank(abs(x))*((sign(x)+1)/2))
  W[i]<-(U[i]-(n*(n+1))/4)
  V[i]<-sqrt(n*(n+1)*((2*n)+1)/24)
  M[i]<-W[i]/V[i]
}
count<-sum(abs(M)>qnorm(1-(a/2)))
prob<-count/m
prob
```

**(4)** To draw the Power Curves of Student's t-test and Wilcoxon Signed Rank Test (n=30,$\gamma = 0.92$):

```r
x<-c(-0.8,-0.6,-0.4,-0.2,0,0.2,0.4,0.6,0.8)
y1<-c(0.8108,0.6372,0.4334,0.1386,0.03968,0.1380,0.4884,0.7754,0.8371)
y2<-c(0.9754,0.8598,0.4715,0.1685,0.04953,0.1706,0.4822,0.8448,0.9752)
matplot(x,cbind(y1,y2),type="l",lty=1,col=c("red","blue"),xlab="Mixing Parameter"
        ,ylab="Powers",main="Multiple Power Curves")

```

**(5)** To find the Empirical (Asymptotic) Cut-off of the t-test:

```
m<-100000
n<-100
g<-0.80
mu<-0
s<-1
T<-double(n)
r<-runif(n)
sum(r<=g)
x<-c()
y<-c()
for(i in 1:m)
{
  for(j in 1:n)
  {
    if(r[j]<=g)
      x[j]<-rnorm(1,mu,s)
    else
      x[j]<-rcauchy(1,mu,s)
  }

  T[i]<-(sqrt(n)*(mean(x)))/sd(x)
}
y<-sort(T)
k<-(m*0.95)
emp_cutoff<-y[k]
emp_cutoff
```

**(6)** To find the Empirical level and Power (Asymptotic) of the t-test statistic:

```
m<-20000
n<-70
g<-0.90
mu<-0
s<-1
T<-double(n)
r<-runif(n)
sum(r<=g)
x<-c()

for(i in 1:m)
{
  for(j in 1:n)
  {
    if(r[j]<=g)
      x[j]<-rnorm(1,mu,s)
    else
      x[j]<-rcauchy(1,mu,s)
  }

  T[i]<-(sqrt(n)*(mean(x)))/sd(x)
}
count<-sum(abs(T)>1.570925692)
power<-count/m
power
```

**(7)** The Empirical level and Power (Asymptotic) of the Robust Statistic:

```r
m<-20000
n<-70
g<-0.90
mu<-0
s<-1
T<-double(n)
r<-runif(n)
sum(r<=g)
x<-c()
for(i in 1:m){
  for(j in 1:n){
    if(r[j]<=g)
      x[j]<-rnorm(1,mu,s)
    else
      x[j]<-rcauchy(1,mu,s)}
  T[i]<-(sqrt(n)*(median(x)))/(IQR(x)/2)
}
count<-sum(abs(T)>3.158910462)
power<-count/m
power
```

**(8)** The q-q plot of the t-test(empirical) statistic:

```r
m<-100000
n<-200
g<-0.94
mu<-0
s<-1
T<-double(n)
r<-runif(n)
sum(r<=g)
x<-c()
y<-c()
for(i in 1:m)
{
  for(j in 1:n)
  {
    if(r[j]<=g)
      x[j]<-rnorm(1,mu,s)
    else
      x[j]<-rcauchy(1,mu,s)
  }

  T[i]<-(sqrt(n)*(mean(x)))/sd(x)
}
l<-(T-mean(T))/sd(T)
qqplot(l,rnorm(m),xlab='Statistic',ylab='Normal')
```

**(9)** The q-q plot of the Robust Statistic:

```r
m<-100000
n<-200
g<-0.94
mu<-0
s<-1
T<-double(n)
r<-runif(n)
sum(r<=g)
x<-c()
for(i in 1:m)
{
  for(j in 1:n)
  {
    if(r[j]<=g)
      x[j]<-rnorm(1,mu,s)
    else
      x[j]<-rcauchy(1,mu,s)
  }

  T[i]<-(sqrt(n)*(median(x)))/(IQR(x)/2)
}
l<-(T-mean(T))/sd(T)
qqplot(l,rnorm(m),xlab='Statistic',ylab='Normal')
```

**(10)** To draw the Power Curves of the New tests Based on empirical t-test statistic and the Robust Statistic (n=30,$\gamma = 0.92$):

```r
x<-c(-0.8,-0.6,-0.4,-0.2,0,0.2,0.4,0.6,0.8)
y1<-c(0.85863,0.8643,0.59262,0.26703,0.11103,0.27978,0.60049,0.82549,0.87475)
y2<-c(0.94939,0.81545,0.54283,0.23925,0.10793,0.23344,0.54195,0.82204,0.95124)
matplot(x,cbind(y1,y2),type="l",lty=1,col=c("red","blue"),xlab="Mixing Parameter"
        ,ylab="Powers",main="Multiple Power Curves")

```

**(11)** Estimating the mixing parameter using EM Algorithm ($\gamma$=0.98):

```r
m<-100000
n<-10
g<-0.98
r<-runif(n)
sum(r<=g)
x<-c()
for(i in 1:m)
{for(j in 1:n)
  {if(r[j]<=g)
      x[j]<-rnorm(1,0,1)
    else
      x[j]<-rcauchy(1,0,1)
  }
}
y<-Mclust(x,G=2)
estimate<-y$parameters$pro
estimate

```